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Robust Machine Learning–Based Ionospheric Delay Grid Modeling for PPP-RTK Under Sparse Reference Stations and Severe Geomagnetic Disturbance

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Key Points:

- A machine learning-based ionospheric delay grid modeling method is proposed to improve PPP-RTK performance
- The method significantly improves ionospheric modeling accuracy in sparse reference station networks compared with the traditional method
- It remains robust during severe geomagnetic disturbances and enables faster PPP-RTK convergence

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Abstract Sparse reference station networks and severe geomagnetic disturbances can significantly degrade the accuracy of regional ionospheric delay modeling at the network side, thereby reducing the positioning performance of PPP-RTK. To address this issue, this study proposes a machine learning-based ionospheric delay grid modeling method that finely models the ionospheric delay computation and interpolation errors embedded in the ionospheric delay residuals. To evaluate the performance of the proposed approach, 274 stations uniformly distributed across Europe were used for experiments based on observation data from 2024. Five machine learning models, including GBDT, XGBoost, RF, SVM, and LSTM, were comparatively analyzed. For a representative test station (TLL1) with an average reference station spacing of 260 km, the ionospheric delay products generated by the proposed method exhibit errors smaller than 10 cm for most epochs. Compared with the traditional inverse distance weighting (IDW) method, the five machine learning models improve the average root mean square error (RMSE) by 60.2%, 57.1%, 61.6%, 49.3%, and 50.7%, respectively. During severe geomagnetic disturbances, the probability that the modeling error remains smaller than 5 cm exceeds 85%, whereas it is only about 60% when using the IDW method. PPP-RTK experiments further demonstrate that the proposed method enables rapid or even instantaneous convergence in sparse reference station networks. For the TLL1 station, the maximum RMSE in the U direction is reduced by more than 60%, and the average convergence time decreases by up to 75%. These results demonstrate the effectiveness and robustness of the proposed method for enhancing PPP-RTK performance in challenging environments.

Plain Language Summary Accurate ionospheric delay modeling is essential for high-precision PPP-RTK positioning. However, in regions with sparse reference stations or during severe geomagnetic disturbances, traditional modeling methods often lose accuracy and degrade positioning performance. To address this problem, this study proposes a machine learning-based ionospheric delay grid modeling method that learns and corrects errors contained in ionospheric delay residuals. Experiments using GNSS data from 274 stations across Europe show that the proposed method significantly improves modeling accuracy compared with the traditional inverse distance weighting method, even when the average reference station spacing reaches 260 km. The results also demonstrate that the method maintains high reliability during severe geomagnetic disturbances and can greatly improve PPP-RTK positioning performance, enabling faster convergence and higher positioning accuracy in challenging environments.

1. Introduction

The Global Navigation Satellite System (GNSS) has become an indispensable technology in a wide range of fields, including geodesy, disaster early warning, and meteorological research (Dow et al., 2009; Huang et al., 2023; Li et al., 2015). Among various GNSS-based positioning techniques, Precise Point Positioning - Real-Time Kinematic (PPP-RTK) has attracted considerable attention since its introduction (Teunissen et al., 2010; Wübbena et al., 2005). This technology eliminates the limitation of nearby reference stations, enabling users to achieve rapid convergence to centimeter-level positioning accuracy with a single receiver (Chen et al., 2025; Zhang et al., 2011).

The accuracy and reliability of ionospheric delay products are crucial for PPP-RTK positioning. Currently, the accuracy of the global ionospheric maps (GIM) provided by the International GNSS Service (IGS) is about 3.0

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TECUs (Liu et al., 2021), which is insufficient to support high-performance PPP-RTK applications. Therefore, regional ionospheric delay products are typically generated using uniformly distributed reference stations. At the network side, each reference station independently estimates the slant ionospheric delay for all visible satellites. Assuming that the ionospheric delay varies smoothly within a limited region, interpolation models can then be applied to derive the ionospheric delay products for users (Odijk et al., 2016). Common interpolation methods include inverse distance weighting (Yao et al., 2020), low-order surface fitting (Cui et al., 2022), and Kriging interpolation (Hernández-Pajares et al., 2020). Previous studies have demonstrated that these approaches can achieve comparable accuracy in small regional networks (He et al., 2025). In addition, several improved methods have been proposed in recent years, such as the low-degree polynomial with regular grid residuals approach (Yin et al., 2022), slant ionospheric delay models that account for gradient and vertical delay effects (Yang et al., 2026), and radial basis function interpolation (Lyu et al., 2024). However, in regions with sparsely distributed reference stations, the performance of ionospheric delay products generated by these methods degrades significantly, making it difficult to satisfy the accuracy requirements of PPP-RTK positioning.

To address the challenge of constructing PPP-RTK ionospheric delay models in regions with sparse reference stations, extensive research efforts have been made. Gu et al. (2022) proposed a quasi-four-dimensional ionospheric modeling approach to provide wide-area high-precision ionospheric delay corrections (Gu et al., 2022). Cui et al. developed a layered enhancement model that combines a fitting model with a regional interpolation model for wide-area PPP-RTK positioning (Cui et al., 2023). Li et al. introduced a grid-based ionospheric delay weighting method to improve the performance of ionospheric delay products across different spatial scales (Li et al., 2023). Zhang et al. investigated an ionospheric correction generation method based on the best linear unbiased prediction framework and validated its performance in a large-scale network (Zhang et al., 2024). In addition, Du et al. explored the contribution of a triple-frequency multi-constellation computation strategy to large-scale PPP-RTK applications (Du et al., 2025). Despite these efforts, the challenge of constructing accurate PPP-RTK ionospheric delay models in sparsely distributed reference station regions remains unresolved, and further research is still required.

In addition, during extreme space weather events such as severe geomagnetic disturbances, the accuracy and stability of regional ionospheric delay modeling can be significantly degraded (Zhang et al., 2022). Darugna et al. (2021) evaluated the performance of several widely used ionospheric delay interpolation methods, including inverse distance weighting (IDW), ordinary Kriging, and weighted least squares (WLS), under disturbed ionospheric conditions, and further proposed an improved WLS approach that accounts for atmospheric physical behavior during such events (Darugna et al., 2021). Li et al. developed an ionospheric weighting model that considers the relationship between ionospheric activity and correction accuracy, enabling a more appropriate characterization of the uncertainty in ionospheric corrections (Li et al., 2025). To mitigate the degradation of ionospheric correction accuracy during disturbances, Lyu et al. proposed a satellite-specific ionospheric residual integrity monitoring (IRIM) approach to quantify the uncertainty associated with ionospheric corrections (Lyu et al., 2026). Nevertheless, the challenge of maintaining accurate and reliable ionospheric delay modeling under severe geomagnetic disturbances still requires further investigation.

With the rapid development of artificial intelligence, machine learning approaches have provided new opportunities for ionospheric delay modeling, particularly under challenging conditions such as sparse reference station networks and severe geomagnetic disturbances. Various machine learning techniques, including Gradient Boosting Decision Trees (GBDT) (Friedman, 2001), eXtreme Gradient Boosting (XGBoost) (Chen & Guestrin, 2016), Random Forests (RF) (Breiman, 2001), Support Vector Machines (SVM) (Cortes & Vapnik, 1995), and Long Short-Term Memory networks (LSTM) (Hochreiter & Schmidhuber, 1997), have been widely applied to ionosphere-related modeling and prediction tasks. For example, Han et al. employed several machine learning models, including GBDT, to predict the total electron content (TEC) of GPS satellites during periods of high solar activity and geomagnetic storms (Han et al., 2022). Zhukov et al. (2021) developed a global TEC model based on the XGBoost algorithm, achieving a root mean square (RMS) prediction accuracy of 2.5 TECU (Zhukov et al., 2021). Shidler et al. utilized the RF model to investigate quiet-time vertical plasma drifts in the low-latitude F-region ionosphere (Shidler & Rodrigues, 2020). Chen et al. proposed an SVM-based method for forecasting the ionospheric critical frequency 1 hr in advance and validated its performance under different solar activity

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conditions (Chen et al., 2010). In addition, Srivani et al. applied an LSTM model to predict GPS ionospheric delays and achieved prediction errors of 1–2 TECU under both geomagnetically quiet and disturbed conditions (Srivani et al., 2019). These studies demonstrate the considerable potential of machine learning methods for regional ionospheric delay modeling.

To address the accuracy degradation of traditional regional ionospheric delay modeling under sparse reference station networks and strong disturbance conditions, this study proposes a machine learning–based PPP-RTK ionospheric delay grid modeling method. By integrating multi-source ionosphere-related features, the proposed approach enables high-precision modeling and prediction of both model errors and computation errors embedded in ionospheric delay residuals. The remainder of this paper is organized as follows. Section 2 describes the proposed methodology. Section 3 introduces the experimental data and computational strategy. Section 4 presents and discusses the results, and the final section summarizes the main conclusions.

2. Methodology

2.1. PPP-RTK

PPP-RTK consists of a network side and a user side. The network side consists of a set of evenly distributed reference stations and is responsible for generating and broadcasting products such as satellite orbits, satellite clock corrections, pseudorange and phase Observable-Specific Bias (OSB), regional slant ionospheric delay products, and tropospheric delay products. The GNSS raw observation equations on the network side are expressed as follows:

$$\begin{cases} P_{r,i}^{s} = \rho_r^s + c(t_r - t^s) + M_r \cdot T_r + \gamma_i \cdot I_{r,1}^{s} + d_{r,i} - d_i^s + \varepsilon_{P_{r,i}^s} \\ L_{r,i}^{s} = \rho_r^s + c(t_r - t^s) + M_r \cdot T_r - \gamma_i \cdot I_{r,1}^{s} + \lambda_i\left(N_{r,i}^{s} + b_{r,i} - b_i^s\right) + \varepsilon_{L_{r,i}^s} \end{cases}, \tag{1}$$

which $P_{r,i}^{s}$ and $L_{r,i}^{s}$ denote pseudorange and phase observations. The superscript s denotes the satellite, and the subscripts r and i denote the receiver and the frequency, respectively. ρ_r^s is the geometric distance between the receiver r and the satellite s. c is the speed of light in vacuum. t_r and t^s denote receiver and satellite clock offset. M_r represents the mapping function in the zenith direction of the troposphere delay. T_r is the tropospheric delay in the zenith direction. T_r includes the tropospheric hydrostatic delay and the tropospheric wet delay. It is generally assumed that the hydrostatic delay can be accurately modeled using the Saastamoinen model (Saastamoinen, 1972), while the wet delay, which is affected by water vapor variations, needs to be precisely estimated. γ_i is the slant ionospheric delay coefficient associated with frequency i. $I_{r,1}^{s}$ denotes the first-order ionospheric delay on the slant path for the first frequency. $d_{r,i}$ and d_i^s are the pseudorange hardware delays at the receiver side and the satellite side, respectively. λ_i is the wave length in the frequency i. $N_{r,i}^{s}$ denotes the phase ambiguity in the frequency i. $b_{r,i}$ and b_i^s are the phase hardware delays at the receiver side and the satellite side. $\varepsilon_{P_{r,i}^s}$ and $\varepsilon_{L_{r,i}^s}$ represent the residual errors of the pseudorange and phase measurements, respectively, mainly including observation noise, multipath errors, and other modelable error terms. The parameters in Equation 1 exhibit significant correlations, causing the functional model to be rank-deficient. To solve this equation and eliminate the rank deficiency, the parameters need to be reparameterized, as follow:

$$\begin{cases} P_{r,i}^{s} = \rho_r^s + c \cdot \tilde{t}_r + M_r^w \cdot Z_r + \gamma_i \cdot \tilde{I}_{r,1}^{s} + \varepsilon_{P_{r,i}^s} \\ L_{r,i}^{s} = \rho_r^s + c \cdot \tilde{t}_r + M_r^w \cdot Z_r - \gamma_i \cdot \tilde{I}_{r,1}^{s} + \lambda_i \cdot \tilde{N}_{r,i}^{s} + \varepsilon_{L_{r,i}^s} \end{cases}, \tag{2}$$

in which M_r^w and Z_r denote the wet tropospheric mapping function and the zenith wet delay, respectively. $\tilde{t}_r$, $\tilde{I}_{r,1}^{s}$, and $\tilde{N}_{r,i}^{s}$ represent the reparameterized receiver clock offset, satellite clock offset, slant ionospheric delay, and phase ambiguity, respectively, which are expressed as follows:

$$\begin{cases} \tilde{t}_r = t_r + \dfrac{1}{c}(\alpha \cdot d_{r,1} - \beta \cdot d_{r,2}) \\ \tilde{I}^s_{r,1} = I^s_{r,1} - \beta\left(d_{r,1} - d_{r,2} - d^s_1 + d^s_2\right) \\ \tilde{N}^s_{r,1} = N^s_{r,1} + b_{r,1} - b^s_1 - \dfrac{1}{\lambda_1}\left[(\alpha+\beta)\left(d_{r,1} - d^s_1\right) - 2\beta\left(d_{r,2} - d^s_2\right)\right] \\ \tilde{N}^s_{r,2} = N^s_{r,2} + b_{r,2} - b^s_2 - \dfrac{1}{\lambda_2}\left[2\alpha\left(d_{r,1} - d^s_1\right) - (\alpha+\beta)\left(d_{r,2} - d^s_2\right)\right] \end{cases}, \tag{3}$$

where the hardware delay d^s_1, d^s_2, b^s_1 and b^s_2 can be corrected using OSB products. α and β denote the coefficients, which are specifically expressed as follows:

$$\begin{cases} \alpha = \dfrac{\gamma_2}{\gamma_2 - 1} \\ \beta = \dfrac{1}{\gamma_2 - 1} \end{cases}. \tag{4}$$

Similarly, the satellite clock and orbit can be obtained from the corresponding products. The coordinates of the reference stations are usually known. After removing the modelable errors, such as solid Earth tides, antenna phase center variations, Earth rotation correction, and relativistic effects, the atmospheric delay parameters to be estimated on the regional reference station network side can be expressed as follows:

$$X = \left[\tilde{t}_r \quad Z_r \quad \tilde{I}^s_{r,1} \quad \tilde{N}^s_{r,1} \quad \tilde{N}^s_{r,2}\right]. \tag{5}$$

These parameters can be estimated using Kalman filtering or sequential least squares. The estimated ambiguities, having absorbed hardware biases and other effects, are real-valued rather than integer, and therefore ambiguity fixing is required. The original ambiguities are combined to form wide-lane and narrow-lane ambiguities. The wide-lane ambiguities are fixed using the decision function strategy (Blewitt, 2012), while the narrow-lane ambiguities are fixed using the least squares ambiguity decorrelation adjustment (LAMBDA) algorithm in this study (Teunissen, 1995). After the ambiguities are fixed, they are added to the original equations as virtual observation equations, and a refiltering process is performed to obtain the ambiguity fixed ionospheric delay and tropospheric delay.

After the regional reference stations compute the slant ionospheric and tropospheric delays, these products are broadcast to the users. Based on their geographic distribution and other information, users can apply interpolation models to derive the slant ionospheric and tropospheric delay products at the user end. Since the computed slant ionospheric delays contain part of the receiver side hardware biases and it is difficult to precisely separate these biases, the satellite with the highest elevation angle is selected as the reference satellite in this study, and single differenced slant ionospheric delay constraints are applied at the user end to eliminate the impact of these biases. Then, the functional model on the user side can be expressed as follows:

$$\begin{cases} \tilde{P}^s_{r,i} = \nu^s_r \cdot x_r + c \cdot \tilde{t}_r + M^w_r \cdot Z_r + \gamma_i \cdot \tilde{I}^s_{r,1} + \varepsilon_{P^s_{r,i}} \\ \tilde{L}^s_{r,i} = \nu^s_r \cdot x_r + c \cdot \tilde{t}_r + M^w_r \cdot Z_r - \gamma_i \cdot \tilde{I}^s_{r,1} + \lambda_i \cdot \tilde{N}^s_{r,i} + \varepsilon_{L^s_{r,i}} \\ \hat{Z}_r - Z_{r,0} = Z_r + \varepsilon_{\hat{Z}_r} \\ \left(\hat{I}^s_{r,1} - \hat{I}^k_{r,1}\right) - \left(\tilde{I}^s_{r,1,0} - \tilde{I}^k_{r,1,0}\right) = \tilde{I}^s_{r,1} - \tilde{I}^k_{r,1} + \varepsilon_I \end{cases}, \tag{6}$$

in which $\tilde{P}^s_{r,i}$ and $\tilde{L}^s_{r,i}$ are the pseudorange and phase observations after removing the modelable errors, respectively. ν^s_r denotes the coefficient matrix. x_r is the receiver position to be estimated. $\hat{Z}_r$ and $Z_{r,0}$ denote the tropospheric delay product and user side tropospheric delay initial value, respectively. $\hat{I}^s_{r,1}$ and $\hat{I}^k_{r,1}$ are the

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regional slant ionospheric delay products of satellite s and the reference satellite k, respectively. $\tilde{I}^{s}_{r,1,0}$ and $\tilde{I}^{k}_{r,1,0}$ are the ionospheric delay initial value. $\varepsilon_{\tilde{Z}_r}$ and ε_I represent the errors of the products.

2.2. Machine Learning-Based Ionospheric Delay Grid Modeling

High-precision regional ionospheric delay modeling is crucial for PPP-RTK user applications, as it can effectively accelerate convergence and improve positioning accuracy. However, in regions with sparse reference stations or during strong geomagnetic disturbances, the spatial and temporal correlations of the ionosphere between reference stations are significantly weakened. As a result, the accuracy of regional slant ionospheric delay modeling based on traditional numerical interpolation methods deteriorates, making them insufficient to support high-performance PPP-RTK applications. In addition, although most satellite- and receiver-related hardware biases can be largely eliminated through between-satellite single-differencing and OSB, the slant ionospheric delays estimated at reference stations still contain residual errors arising from computational inaccuracies and imperfect modeling. These residual components, together with interpolation errors, are difficult to characterize using conventional mathematical models. If such errors can be effectively captured and mitigated, the performance of slant ionospheric delay modeling in sparse reference station networks or under disturbed ionospheric conditions could be significantly improved. However, due to the highly complex and dynamic nature of the ionosphere and the numerous factors influencing ionospheric delays, it is difficult to accurately model these errors using conventional mathematical approaches. With the rapid advancement of artificial intelligence, machine learning techniques provide new opportunities for improving slant ionospheric delay modeling.

The errors of the regional slant ionospheric delay modeling can be refined and expressed as follows:

$$\varepsilon_I = \varepsilon_{I_0} + \varepsilon_m + \varepsilon_r, \tag{7}$$

in which ε_{I_0} denotes ionospheric delay estimation error. ε_m denotes the model error. ε_r denotes random noise, which is inevitable. The estimation error of the slant ionospheric delay is influenced by the noise affecting the pseudorange and carrier-phase observations at reference stations, as well as by factors including the signal-to-noise ratio, multipath effects, satellite geometry, the level of ionospheric activity, geomagnetic field conditions, and solar radiation intensity. Because these three types of errors are intertwined, it is difficult to achieve accurate separation and modeling using purely mathematical models. Data-driven machine learning approaches are expected to enable effective modeling of both slant ionospheric delay computation errors and model errors, thereby enhancing the performance of ionospheric delay grid modeling for PPP-RTK in sparse reference station networks and under severe geomagnetic disturbances.

Commonly used machine learning methods mainly include GBDT, XGBoost, RF, SVM, and LSTM. Therefore, in this study, these commonly used machine learning models are employed to learn the slant ionospheric delay residuals computed at the network reference stations, thereby enabling the modeling of errors in slant ionospheric delay estimation as well as interpolation model errors. Through modeling both the slant ionospheric delay estimation errors and the inherent model deficiencies, a machine learning-based grid model for PPP-RTK ionospheric delay is developed. The specific procedure of this method is illustrated in Figure 1.

First, a subset of stations is randomly selected from the reference station network to act as rover stations. The GNSS raw observation data from all stations are processed using the undifferenced and uncombined PPP algorithm to estimate the slant ionospheric delays, their uncertainties, and other characteristic parameters at both the reference and rover stations. Subsequently, the slant ionospheric delay at the rover station is interpolated from the reference stations using a local ionospheric interpolation model. The ionospheric delay obtained from the undifferenced and uncombined PPP solution at the rover station is regarded as the reference value, and the corresponding ionospheric delay residuals are then derived. Next, feature parameters related to slant ionospheric delay estimation errors are extracted. These features include satellite position, elevation angle, and azimuth angle to characterize the spatial distribution of satellites. The rate of TEC index (ROTI) is introduced to quantify the level of ionospheric disturbances affecting signal propagation. The mathematical expression of ROTI is expressed as below (Luo et al., 2022):

$$\mathrm{ROTI} = \sqrt{< \mathrm{ROT}^2 > - < \mathrm{ROT} >^2}, \tag{8}$$

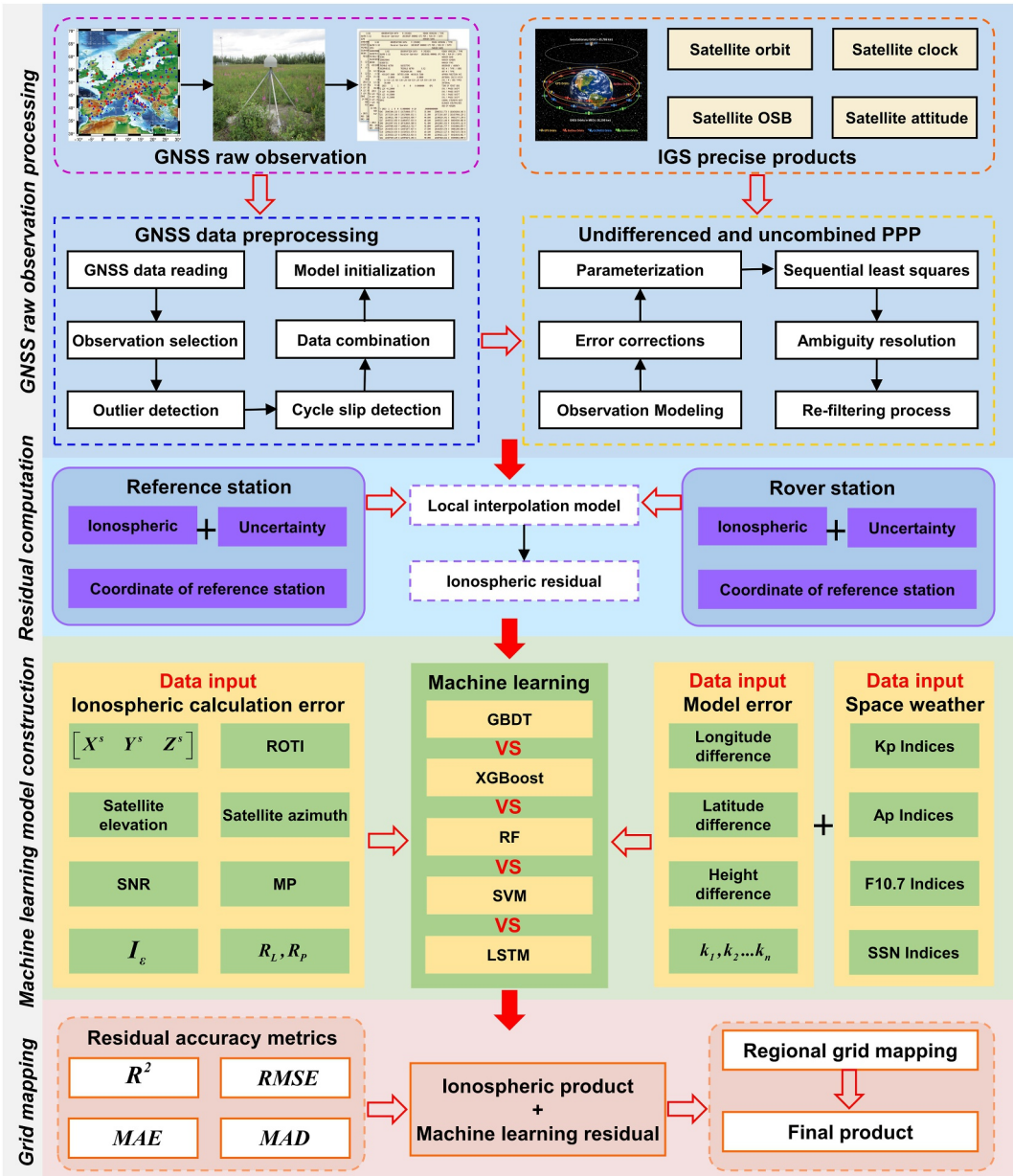


Figure 1. Workflow of PPP-RTK ionospheric delay grid modeling based on machine learning.

where the angle brackets represent the average value in a period of time. ROT represents the change rate of TEC and can be expressed as follows:

$$\begin{cases} \mathrm{ROT} = \dfrac{\mathrm{STEC}_k^s - \mathrm{STEC}_{k-1}^s}{t_k - t_{k-1}} \\ \mathrm{STEC} = \dfrac{c}{K}\left(\dfrac{\Phi_1}{f_1} - \dfrac{\Phi_2}{f_2}\right) \cdot \dfrac{f_1^2 \cdot f_2^2}{f_1^2 - f_2^2} \end{cases}, \tag{9}$$

where k represents the epoch time. f_1 and f_2 represent the carrier frequency of satellite signals. Φ_1 and Φ_2 represent the carrier phase measurements in cycle. K denotes the ionospheric constant, which is typically taken as 40.3 m^3/s^2 c is the speed of light in vacuum. In addition, the signal-to-noise ratio (SNR) and multipath (MP) indicators are incorporated to describe the observation environment and data quality. Furthermore, the

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Table 1
The Comparison of Multiple Machine Learning Models

Model	Nonlinear modeling capability	Adaptability to small-sample scenarios	Computational efficiency
GBDT	strong	good	moderate
XGBoost	very strong	very good	high
RF	strong	very good	high
SVM	strong	excellent	low
LSTM	very strong	poor	low

ionospheric formal error, together with pseudorange and carrier-phase residuals, is used to represent the precision of slant ionospheric delay estimation. These features provide information on satellite–receiver geometry, ionospheric activity, observation quality, the uncertainty, and residual characteristics of the estimated ionospheric delays. Meanwhile, feature parameters related to interpolation model errors are constructed according to the spatial configuration of the reference station network. These mainly include the differences in longitude, latitude, and elevation between the reference and rover stations, which describe their relative spatial relationships, as well as the weighting factors $k_1, k_2, ...k_n$ that represent the contribution of each reference station to the ionospheric modeling at the rover location. These features characterize the spatial relationship between the target and reference stations and provide information on the spatial variability of ionospheric delays. In addition, to enhance the model performance under different levels of geomagnetic disturbance, several space weather indices are incorporated as feature parameters in the ionospheric delay modeling. These include the Kp and Ap indices, which characterize geomagnetic activity, as well as the F10.7 index and sunspot number, which reflect solar activity conditions. These features were jointly introduced into the machine learning model to provide complementary information from GNSS observations, spatial interpolation, and space-weather conditions. It should be noted that their relative contributions are not necessarily constant, as they may vary with the ionospheric activity, geographical region, satellite geometry, reference-station distribution, and observation period.

Subsequently, long-term feature data sets are used to train the machine learning models, with the ionospheric delay residual time series serving as the label data for residual modeling. Since the performance of machine learning models may vary across different regions and time periods, multiple algorithms, including GBDT, XGBoost, RF, SVM, and LSTM, are employed for training and comparison to investigate their applicability to ionospheric delay residual modeling. To ensure a fair comparison, all five machine learning models use the same input features in this study. The comparison of multiple machine learning models is presented in Table 1. GBDT exhibits strong nonlinear modeling capability and good adaptability to small-sample scenarios, with moderate computational efficiency. XGBoost demonstrates very strong nonlinear modeling ability, very good performance in small-sample conditions, and high computational efficiency. RF also shows strong nonlinear modeling capability and very good small-sample adaptability, with high computational efficiency. SVM performs well in terms of nonlinear modeling and achieves excellent adaptability in small-sample and high-dimensional feature spaces; however, its computational efficiency is relatively low. In contrast, LSTM exhibits very strong nonlinear modeling capability but relatively poor adaptability to small-sample conditions. Additionally, it has low computational efficiency.

The performance of each model is evaluated using accuracy metrics such as the coefficient of determination R^2, root mean square error (RMSE), mean absolute error (MAE), and median absolute deviation (MAD). The specific definitions of these evaluation metrics are given as follows:

$$\begin{cases} R^2 = 1 - \dfrac{\sum_{i=1}^{n}\left(y_i - \hat{y}_i\right)^2}{\sum_{i=1}^{n}\left(y_i - \overline{y}_i\right)^2} \\ \mathrm{RMSE} = \sqrt{\dfrac{1}{n}\sum_{i=1}^{n}\left(y_i - \hat{y}_i\right)^2} \\ \mathrm{MAE} = \dfrac{1}{n}\sum_{i=1}^{n}\left|y_i - \hat{y}_i\right| \\ \mathrm{MAD} = \mathrm{median}\left(\left|y_i - \mathrm{median}(y)\right|\right) \end{cases}, \tag{10}$$

where y_i, $\hat{y}_i$, and $\overline{y}_i$ denote the real value, the predicted value, and the mean of the real values, respectively. Real value denotes the reference ionospheric delay obtained from the PPP solution at the rover station. Finally, the machine learning model with the best performance is selected according to the evaluation metrics, and the trained model is applied to real-time ionospheric delay residual modeling. The ionospheric residuals estimated by the

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machine learning model are introduced as compensation terms into the computed slant ionospheric delay products. Subsequently, the compensated ionospheric delay values are interpolated onto the regional grid points using the IDW method and broadcast to the user side as ionospheric delay products. The choice of the projection function depends on several factors, including the terrain characteristics of the study area, the spatial resolution of the regional grid, and the spatial distribution and density of the reference stations. Therefore, different mapping functions may be adopted according to the specific application scenario.

Uncertainty is an important indicator of PPP-RTK ionospheric delay products, as it can be used to evaluate product performance and to adjust the strength of stochastic constraints. The uncertainty representation of the machine learning-based PPP-RTK ionospheric delay grid product proposed in this study is expressed as follows:

$$\sigma_i = k \cdot \sigma_{r,i}, \tag{11}$$

where k denotes the scale factor, and $\sigma_{r,i}$ represents the ionospheric delay uncertainty derived from the interpolation model. The specific value of the scale factor can be determined through statistical analysis based on the actual performance of the machine learning model on the validation data set.

In this study, a residual-learning strategy is adopted, in which machine learning is used to model the ionospheric delay residuals derived from a conventional interpolation method. Compared with directly applying machine learning to model ionospheric delays from slant observations, this hybrid approach offers several advantages. First, the interpolation model provides a physically meaningful baseline, ensuring the fundamental spatial characteristics of the ionosphere are preserved. Second, the machine learning model focuses only on learning the residual errors, which are typically smaller in magnitude and exhibit more regular patterns, thereby reducing model complexity and improving generalization capability. In contrast, a direct machine learning approach would require the model to learn the full ionospheric delay structure, which increases the risk of overfitting and may lead to degraded performance, particularly in sparse reference station networks and under disturbed ionospheric conditions. Therefore, the proposed residual-learning framework can be regarded as an effective combination of model-driven and data-driven approaches, providing improved robustness and accuracy for PPP-RTK applications.

3. Experimental Data and Processing Strategy

To validate the performance of the machine learning-based PPP-RTK ionospheric delay grid modeling method proposed in this study, GNSS observation data from 274 continuously operating reference stations (CORS) uniformly distributed across Europe (https://epncb.oma.be/) are used for performance evaluation. The spatial distribution of these stations is shown in Figure 2. Among them, 70 stations are designated as rover stations (red circles in the figure), while the remaining 204 stations serve as reference stations (blue triangles). The year 2024 falls within Solar Cycle 25, a period characterized by pronounced solar activity, during which the Southern Hemisphere of the Sun exhibited particularly strong activity, with sunspot numbers reaching their peak (Rodríguez et al., 2024). Such enhanced solar activity can significantly affect the performance of regional ionospheric delay modeling methods. Therefore, GNSS observation data from 2024 are selected in this study for modeling the ionospheric delay residuals.

The reference station network employs the undifferenced and uncombined PPP technique to estimate ionospheric delays, their uncertainties, and other characteristic parameters. The detailed data processing strategies and model correction schemes are summarized in Table 2. The PPP-RTK processing was conducted using an in-house software developed based on Position and Navigation Data Analyst (PANDA). Although the rover stations are static, a kinematic processing strategy was adopted, with epoch-wise coordinate estimation under loose constraints, to enable an unbiased assessment of PPP-RTK performance. In the experiments, observations from the GPS and Galileo constellations are used with a sampling interval of 30 s. The ionospheric delay at the rover station is interpolated using the IDW method, while the rover data are processed using the same strategy. Since the estimated ionospheric delay products absorb receiver hardware delays, the satellite with the highest elevation angle is selected as the reference satellite to eliminate this effect. When constructing the data set, the input feature parameters at each epoch correspond to the two satellites involved in the single-difference processing. In this study, the data from 2024 are divided into training and validation sets with a ratio of 8:2, and the trained models

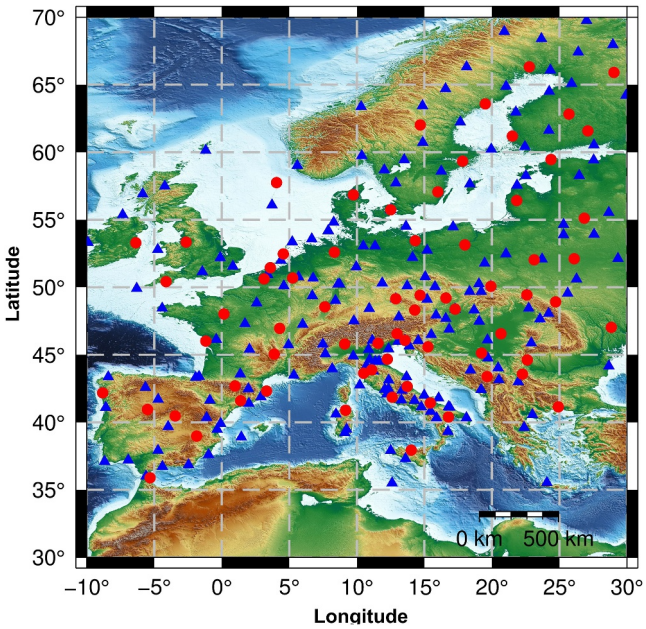


Figure 2. Spatial distribution of the experimental stations.

are then used to predict the ionospheric delay residuals for the data from 2025. Finally, the residuals estimated by the machine learning models are added to the ionospheric delays obtained from the IDW method, and the corrected ionospheric delay products are mapped onto a 0.5° × 0.5° grid.

For the GBDT model, the number of weak learners is set to 300 after multiple trials to balance model complexity and overfitting risk. The learning rate is set to 0.05, which controls the contribution of each tree to the final model. The maximum depth of each tree is set to 3, allowing the model to remain smooth and better handle noisy data. For the XGBoost model, the squared error is adopted as the regression objective function. The learning rate is set to 0.05, and the maximum depth of each tree is set to 8. To mitigate overfitting and introduce randomness, the subsample ratio for each boosting iteration is set to 0.8. In addition, the proportion of features used for each tree is set to 0.7 to reduce feature correlation. For the RF model, the squared error is also used as the regression objective function. To enhance the diversity of individual trees, the sampling ratio of training samples is set to 0.8, and the proportion of features used by each tree is also set to 0.8. Bagging is applied in each iteration, and the maximum number of leaf nodes for each tree is set to 64. For the SVM model, the radial basis function (RBF) is adopted as the kernel function to capture nonlinear relationships in the data. The penalty parameter is set to 10, and the width of the insensitive loss function is set to 0.1. For the LSTM model, the mini-batch gradient descent algorithm is employed to iteratively update the model parameters, improving training stability and computational efficiency. The mean squared error (MSE) is used as the loss function, and the Adam optimizer is applied to minimize the MSE. By adaptively adjusting the learning rate, the Adam optimizer improves the convergence speed and robustness of the model when handling non-stationary time-series data.

Table 2
The Detailed Data Processing Strategies and Model Correction Methods

Item	Strategy
Satellite system	GPS and Galileo
Combination model	undifferenced and uncombined
Sampling rate	30 s
Cutoff elevation angle	7°
Coordinates constrained	Reference station: fix
	Rover station: 100 m loose constraint
Satellite orbit and clock	WUM rapid products
Satellite attitude and OSB	WUM rapid products
Observation prior accuracy	Pseudorange: 0.6 m phase: 0.01 cycle
PCO and PCV correction	IGS20.atx
Parameter estimation method	Sequential least squares
Receiver clock processing	Estimated as white noise
Ionospheric delay processing	Estimated with random walk
Tropospheric delay processing	Estimated with random walk
User side atmospheric constraints	Ionospheric delay: product accuracy
	Tropospheric delay: product accuracy
Ambiguity resolution method	Partial ambiguity resolution
	Wide-lane: decision function
	Narrow-lane: LAMBDA

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Table 3
Distribution of the Station Network

Rover station	Reference station	Maximum inter-station distance (km)	Average inter-station distance (km)	Location
LCRA	UNEN, TOPP, JATE, TRMI	74.3	67.9	Italy
ENTZ	BSCN, PFA3, KARL, DILL	197.0	144.6	France
OLK2	FINS, TUO2, ORIV, SUN6	234.1	155.8	Finland
TLL1	VLN1, SCIL, ADAR, MLHD	375.7	260.2	Estonia

4. Results and Discussion

4.1. Residual Characteristics of Regional Ionospheric Delay

To achieve precise modeling of PPP-RTK ionospheric delay residuals, it is necessary to first analyze their characteristics in detail. In this section, four reference station networks with different inter-station distances are selected for experimental analysis. The corresponding rover stations are LCRA, ENTZ, OLK2, and TLL1, and the detailed distribution of the station networks is summarized in Table 3. Among these networks, the one used for modeling the ionospheric delay at rover station TLL1 has the largest spatial scale, with a maximum inter-station distance of 375.7 km and an average inter-station distance of 260.2 km. Such large inter-station separations pose a considerable challenge to PPP-RTK ionospheric delay modeling.

The GPS and Galileo single-differenced ionospheric delay residuals obtained using the inverse distance weighting (IDW) method at the LCRA, ENTZ, OLK2, and TLL1 stations on 1 January 2024 are shown in Figure 3, where different colors represent different satellites. When the inter-station distance of the reference station network is small, the single-differenced ionospheric delay residuals remain relatively small. However, as the inter-station distance increases, the residuals gradually increase, which becomes the primary source of interpolation model errors. When the average inter-station distance reaches 260.2 km, the RMSE of the GPS ionospheric residuals increases to 8.38 cm, with errors exceeding 10 cm at many epochs. Such large errors would significantly degrade the positioning performance of PPP-RTK user terminals. Although the Galileo satellites exhibit slightly better performance, their RMSE still reaches 5.96 cm, and the residuals during certain periods also exceed 10 cm. These results indicate that traditional interpolation models relying solely on the spatial distribution of reference stations still require further improvement for PPP-RTK applications.

To further analyze the characteristics of regional ionospheric delay residuals, the residual distributions of the single-differenced regional ionospheric delay from four different networks were computed for the period from January 1 to 7, 2024. The residual distributions for satellites G02 and E02 are illustrated in Figure 4, where the red curves denote the fitted normal distributions. At the LCRA station, the single-differenced ionospheric delay residuals of satellite G02 are concentrated around zero and exhibit good agreement with the fitted normal distribution. The mean and standard deviation of the residuals are −0.65 and 0.83 cm, respectively. Most residuals fall within ±2 cm, indicating that they are primarily dominated by random noise. As the inter-station distance increases, the residual distribution becomes progressively more dispersed, and the goodness of fit to the normal distribution gradually deteriorates. This suggests that random noise accounts for only a small portion of the residual series, while interpolation model errors and ionospheric delay estimation errors become the dominant components. Furthermore, both the mean and the standard deviation of the residuals increase significantly with increasing inter-station distance. When the average inter-station distance reaches 260 km, the mean residual of satellite G02 increases to −4.97 cm and the standard deviation rises to 6.87 cm. Under such conditions, the reliability of the regional ionospheric delay products is substantially reduced, providing only limited improvement in accelerating the convergence of PPP-RTK user positioning.

4.2. Performance Validation in Sparse Reference Station Networks

To validate the performance of the proposed machine learning-based PPP-RTK ionospheric delay grid modeling method, GNSS data from European CORS stations throughout 2024 were used to compute the feature parameters and ionospheric delay residuals. The resulting time series were then used as inputs to the machine learning models to construct compensation models for ionospheric delay estimation errors and interpolation model errors. The

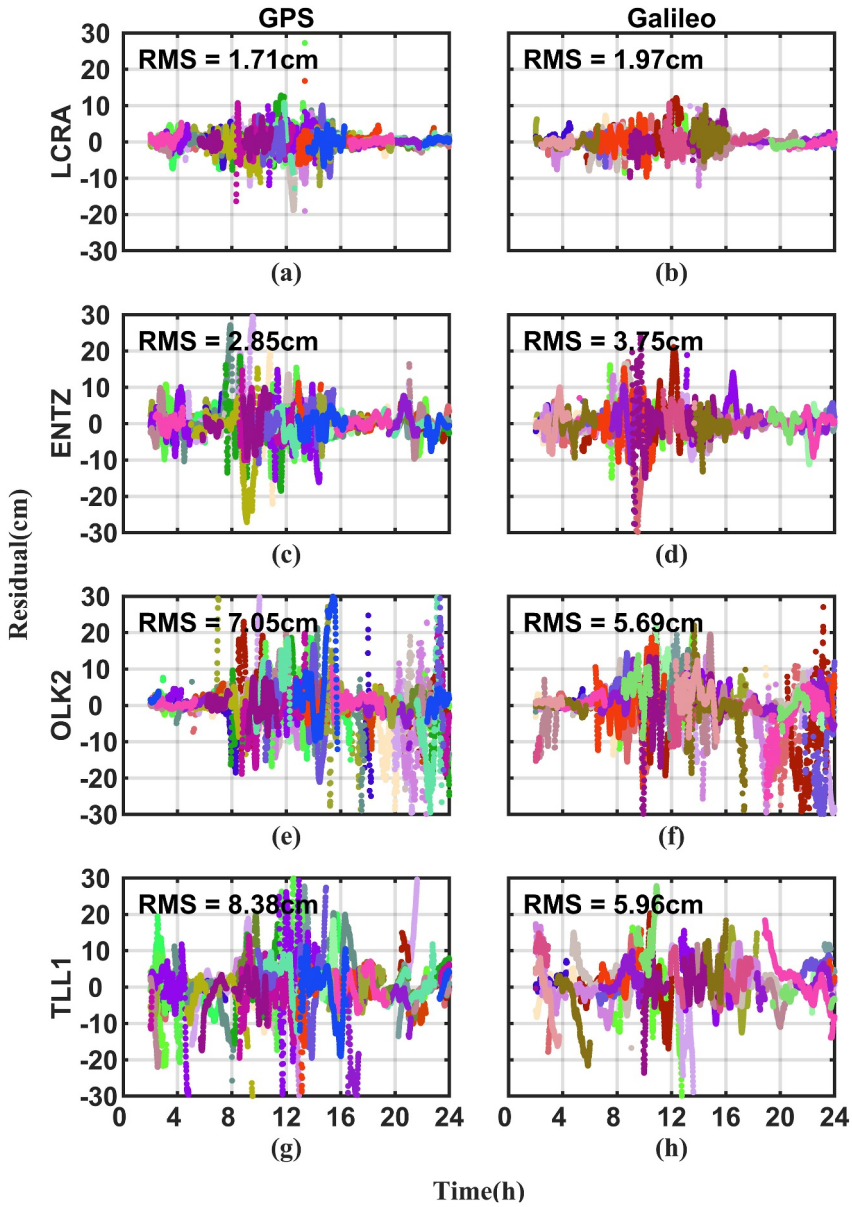


Figure 3. The single-differenced ionospheric delay residuals under different reference station network distributions.

estimated compensation terms were subsequently added to the ionospheric delay products and mapped onto a grid. Figure 5 presents the single-differenced ionospheric delay residuals at the ENTZ and TLL1 stations on day of year (DOY) 001 in 2025 obtained using different machine learning methods, where different colors represent different satellite pairs. When only the IDW method is employed for ionospheric delay interpolation, large errors occur between 8:00 and 16:00 (UTC) at the ENTZ station, where the average inter-station distance is approximately 144 km. This station, located in France, experiences larger errors during this period mainly due to solar illumination. During daytime hours, the ionosphere becomes more active, and interpolation models relying solely on spatial distance tend to produce significant errors. In contrast, the machine learning-based ionospheric delay grid modeling method significantly improves the accuracy. After incorporating the compensation terms estimated by the five machine learning models into the original ionospheric delay products, the residual ionospheric errors at most epochs are reduced to within 10 cm. The corresponding RMSE improvements reach 62.7%, 65.7%, 61.6%, 62.0%, and 62.8%, respectively, clearly demonstrating the effectiveness of the proposed approach. For the TLL1 station, where the average inter-station distance reaches 260 km, a similarly significant performance

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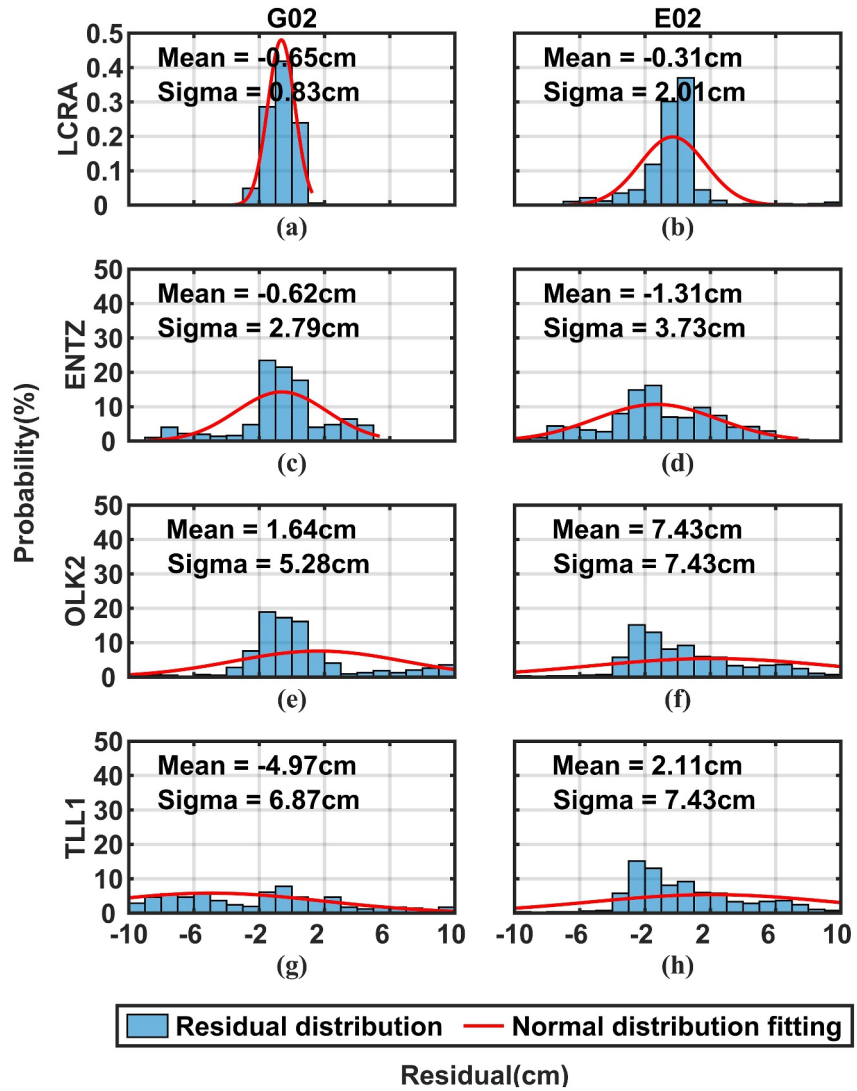


Figure 4. Residual distribution of single-differenced ionospheric delay.

improvement is observed. The ionospheric delay residuals derived using the IDW method exhibit a clear divergence pattern, with an RMSE of 8.72 cm. By contrast, the residual time series obtained using the machine learning-based ionospheric delay grid modeling approach are much more concentrated, with errors at most epochs remaining within 10 cm. The corresponding RMSE improvements are 60.2%, 57.1%, 61.6%, 49.3%, and 50.7%, respectively.

To comprehensively evaluate the performance of the machine learning-based ionospheric delay grid modeling method, the residual distributions and Gaussian fitting results of the single-differenced ionospheric delay on DOY 001 in 2025 at the ENTZ and TLL1 stations were statistically analyzed, as shown in Figure 6. The residuals derived from the IDW method exhibit a wide distribution range with pronounced long-tail characteristics. The standard deviations at the two stations reach 7.50 and 8.72 cm, respectively, indicating large dispersion and limited solution stability. In contrast, the machine learning-based ionospheric delay grid modeling approach significantly narrows the residual distribution and substantially reduces the standard deviation. The residuals become more concentrated and approximately follow an ideal Gaussian distribution. These results suggest that the proposed method can effectively model both ionospheric delay estimation errors and interpolation model errors, leaving predominantly random noise in the remaining residuals. For the IDW method, the mean residuals at the two stations are −0.22 and 0.31 cm, respectively, indicating the presence of systematic biases. By comparison, the residual means obtained from the machine learning-based ionospheric delay model are close to zero,

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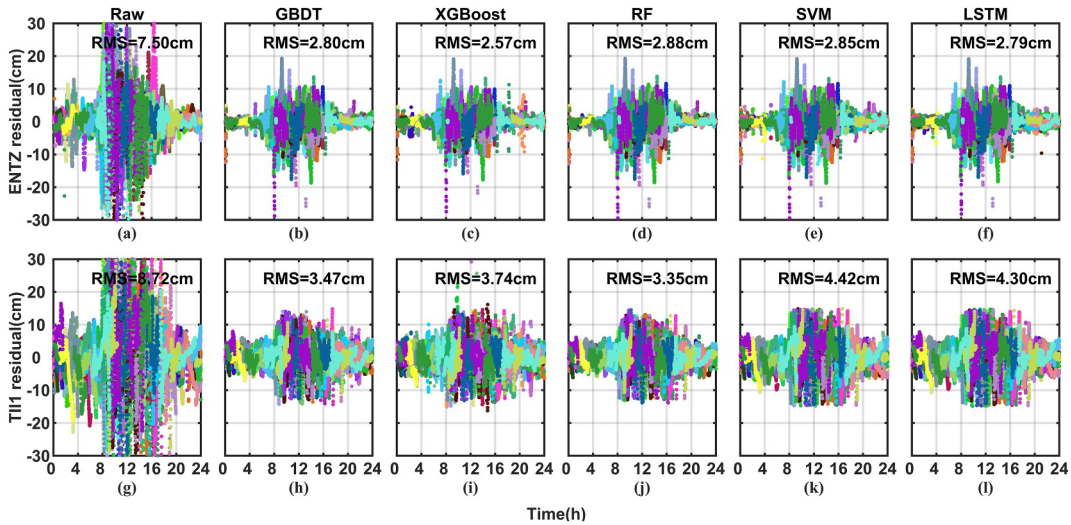


Figure 5. Machine learning-based single-differenced ionospheric delay residuals on DOY 001, 2025.

demonstrating improved unbiasedness. The performance differences among the various machine learning models are relatively small. Among them, XGBoost achieves the best performance at the ENTZ station, whereas RF performs slightly better at the TLL1 station.

Figure 7 presents a comparison between the original ionospheric delay residuals and those predicted by several machine learning models, including GBDT, XGBoost, RF, SVM, and LSTM, for selected satellites at the TLL1 station. For clarity and data alignment, only the first 500 epochs are shown. From an overall perspective, the ionospheric delay residuals derived from the IDW method exhibit pronounced high-noise characteristics and strong non-stationarity, with large-amplitude fluctuations occurring at certain epochs. Significant positive and negative abrupt variations are observed for satellites G12, E02, and E21, with residual magnitudes reaching up to 20 cm. The machine learning models are capable of effectively tracking the variation trends of the ionospheric delay residuals derived from the IDW method, producing smoother and more continuous time series. This indicates that the models successfully capture the dominant variation patterns of the ionospheric delay residuals within the feature space. During periods when the residuals vary slowly, the predictions generated by all models show a high degree of consistency with the original residuals, highlighting their strong ability to capture the underlying trends. However, during epochs characterized by abrupt changes or severe disturbances, notable

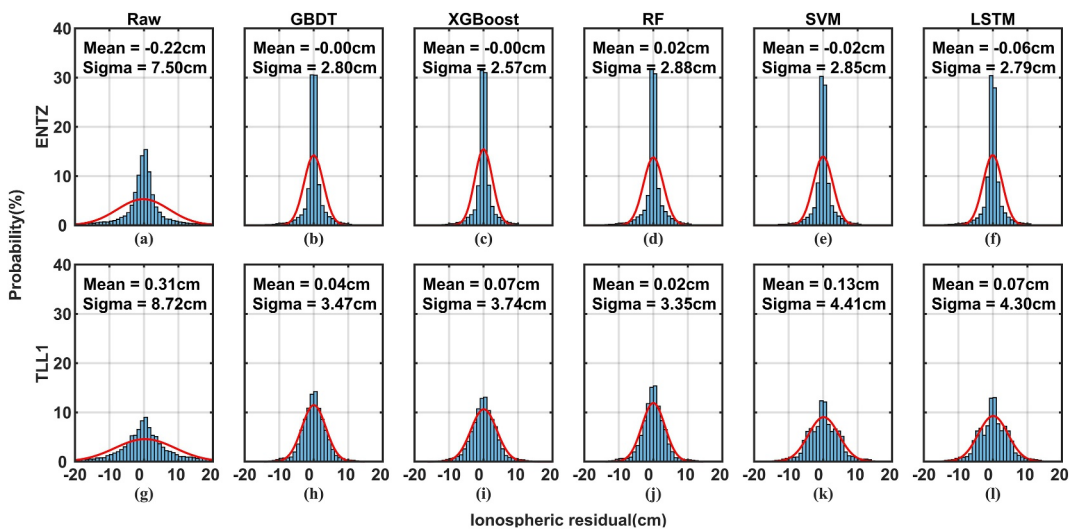


Figure 6. Machine learning-based single-differenced ionospheric delay residual distributions on DOY 001, 2025.

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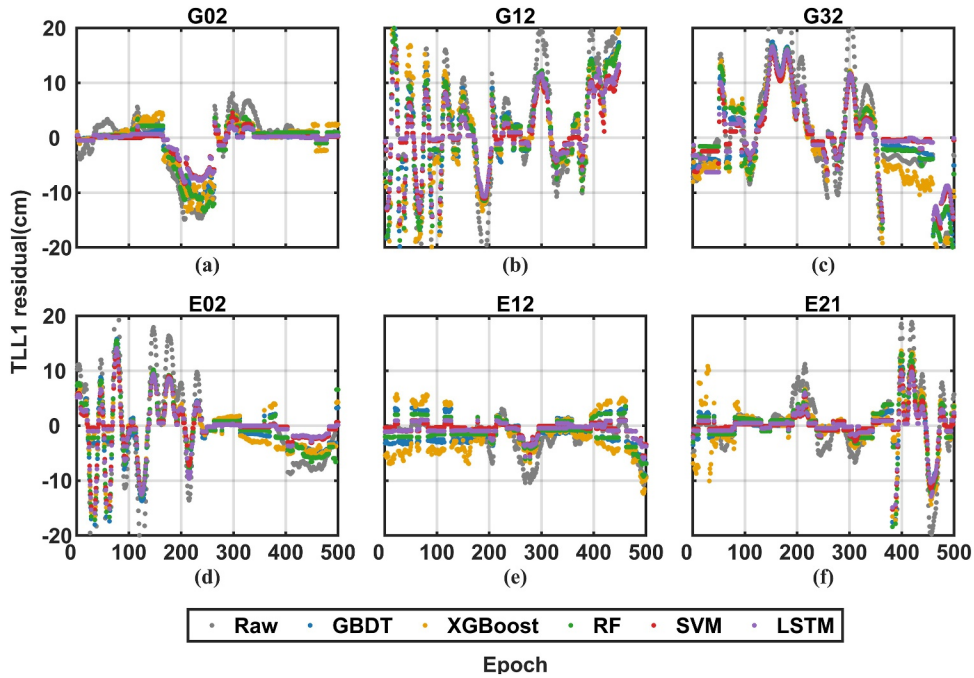


Figure 7. Ionospheric residual modeling based on machine learning for selected satellites.

differences in predictive performance are observed among the models. The GBDT and XGBoost models effectively capture rapid residual variations, maintaining strong consistency with the original residuals and demonstrating superior nonlinear modeling capability and robustness. The RF model exhibits stable performance in representing the overall trend. However, a noticeable smoothing effect appears during high-frequency fluctuation intervals, resulting in attenuated peak amplitudes. The SVM model achieves satisfactory fitting accuracy for residuals of small to moderate magnitude, whereas its prediction errors increase during periods of strong disturbances. The LSTM model effectively characterizes temporal dependencies and produces relatively smooth predictions during continuously varying phases. Nevertheless, a certain degree of lag is observed when responding to abrupt changes.

To comprehensively evaluate the performance of the proposed machine learning-based PPP-RTK ionospheric delay grid modeling method and to mitigate the influence of occasional errors, the regional ionospheric delays for all 70 rover stations on DOY 001 in 2025 were computed using this approach. The interval distribution of the corresponding errors was then statistically analyzed, and the resulting pie chart is presented in Figure 8. As shown in the figure, the ionospheric delay errors derived from the IDW method exhibit a relatively dispersed distribution, with a high proportion of large errors, indicating limited stability. In contrast, the machine learning-based models significantly increase the proportion of small-error samples, and the error distribution clearly shifts toward higher-precision intervals. These results demonstrate that the proposed approach can effectively suppress modeling errors in PPP-RTK ionospheric delay estimation. Specifically, for the traditional spatial interpolation-based regional ionospheric delay method, the proportion of errors exceeding 6 cm reaches 22%. In comparison, for the RF-based method, this proportion is reduced to only 5%, indicating a substantial decrease in the occurrence probability of large errors.

To evaluate the performance of different machine learning models in ionospheric delay residual modeling, five methods were applied to predict the residuals at the LCRA, ENTZ, OLK2, and TLL1 stations from DOY 001 to 007 in 2025. The statistical results of the corresponding accuracy metrics, including the coefficient of determination R^2, RMSE, MAE, and MAD, are presented in Figure 9. From the results, the R^2 values of all stations are greater than 0.5, indicating that all five models are capable of effectively learning the temporal characteristics of the ionospheric delay residuals and exhibit good fitting performance. In the model comparison, the RF method performs best at the TLL1 and OLK2 stations, where the distances between reference stations are large, whereas the XGBoost method performs best at the LCRA and ENTZ stations, where the reference station spacing is small.

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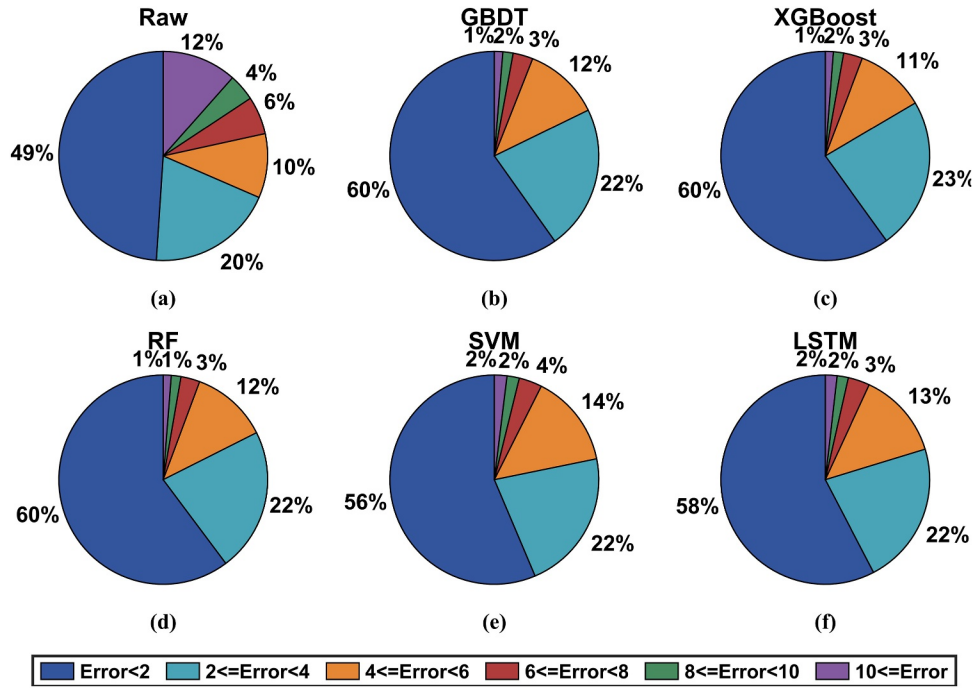


Figure 8. Pie chart of the single-differenced ionospheric delay error distribution for all stations.

The reason for this phenomenon is that when the reference station spacing becomes larger, the spatial correlation of the ionospheric delay weakens and the residuals become more random, making the Random Forest model more robust. In contrast, when the station spacing is smaller and the residuals are smoother, XGBoost can better capture the underlying patterns and thus achieves higher prediction accuracy. RMSE increases with the increase in station spacing, which is consistent with the variation trend of R^2. Using the RF model, the RMSE at the TLL1 station can still be better than 3.5 cm even when the average distance between reference stations reaches 260 km. The variation trend of MAE is generally consistent with that of RMSE and R^2, but its values reflect the average error level more intuitively. The MAE at the LCRA station ranges from about 1.2 to 1.4 cm, whereas at the TLL1 station it increases to more than 3 cm, indicating that the modeling difficulty increases significantly in regions with sparse reference stations. MAD reflects the model's performance in terms of robustness to outliers. The MAD at the LCRA station is the smallest, indicating that the ionospheric residuals at this station are more concentrated with fewer anomalies. In contrast, the MAD at the TLL1 station increases significantly, suggesting that there are more abrupt variations or irregular disturbances in this region.

4.3. Performance Validation of PPP-RTK

To further evaluate the performance of the proposed machine learning-based ionospheric delay grid modeling method in PPP-RTK applications, PPP-RTK experiments were conducted using observation data from the LCRA, ENTZ, OLK2, and TLL1 stations on DOY 001, 2025. To eliminate the influence of tropospheric delay products, the tropospheric delay estimated at each station was introduced as a virtual observation and incorporated into the normal equations. The uncertainty of the ionospheric delay products is critical for user-side positioning. As indicated by the analysis in the previous section, the residuals of the machine learning-based ionospheric augmentation products are significantly reduced, with the average RMSE decreased by approximately 60%. Accordingly, the uncertainty scaling factor is uniformly set to 0.6 in this section. The PPP-RTK positioning results obtained using different ionospheric delay products are presented in Figure 10. For clarity, the positioning results derived from the GBDT, XGBoost, RF, SVM, and LSTM ionospheric delay products are successively shifted upward by 10 cm to avoid overlap. Consequently, the absolute values on the vertical axis do not reflect the true residual magnitudes, but rather their relative variations. Meanwhile, to analyze the convergence characteristics of the positioning solutions and reduce the influence of random effects, the observation data were artificially interrupted every 2 hrs to allow the positioning solutions to reconverge.

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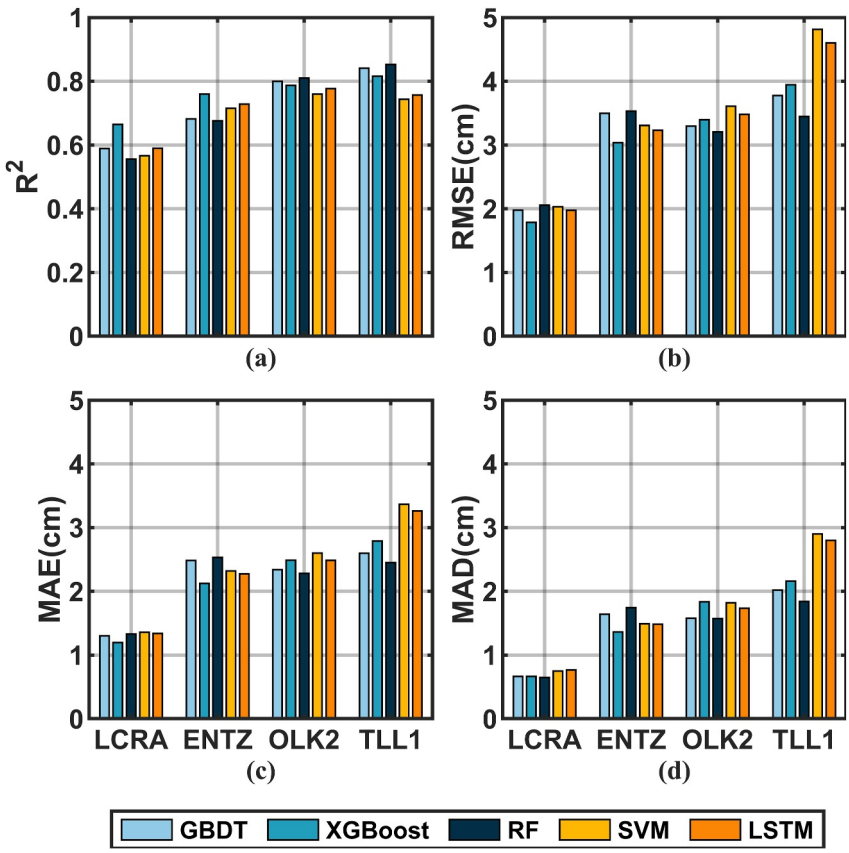


Figure 9. Performance of the five machine learning models under different evaluation metrics.

As shown in Figure 10, for the LCRA station, the PPP-RTK user can achieve rapid or even instantaneous convergence with high positioning accuracy when the IDW method is applied. However, a significant fluctuation appears in the U direction around 10:00, mainly due to the low accuracy of the ionospheric delay products during that period. In contrast, the machine learning-based ionospheric delay grid modeling method significantly improves the positioning performance at that time. As the distance between reference stations increases, the ionospheric delay products derived from the IDW method lead to a noticeable degradation in user positioning performance. The convergence time becomes significantly longer, and large jumps and interruptions appear in the positioning time series. By comparison, the machine learning-based ionospheric delay grid modeling method effectively mitigates this problem and still achieves rapid, or even instantaneous, convergence. The resulting positioning solutions remain continuous and stable, with almost no abrupt jumps or interruptions. These results further confirm the effectiveness and robustness of the machine learning-based PPP-RTK ionospheric delay grid modeling method proposed in this study.

Table 4 presents the maximum positioning errors of the PPP-RTK solutions at different stations using various methods. It can be observed that the raw PPP-RTK solutions exhibit significantly larger maximum errors, particularly at stations with sparse reference networks. This indicates that the low-precision ionospheric delay product severely degrades positioning performance under challenging conditions. After applying different machine learning-based ionospheric modeling methods, the maximum errors are substantially reduced across all stations. In particular, the RF and XGBoost methods demonstrate superior performance, yielding the lowest maximum errors in most cases. For stations with larger inter-station distances, such as OLK2 and TLL1, although all methods still show noticeable errors, the proposed approaches effectively constrain the maximum error growth.

Figure 11 presents the RMSE values in the E, N, and U directions for the LCRA, ENTZ, OLK2, and TLL1 stations on DOY 001 in 2025. These statistics include positioning solutions from epochs that had not yet converged. As the inter-station distance increases, the positioning errors gradually grow, indicating that the ionospheric model

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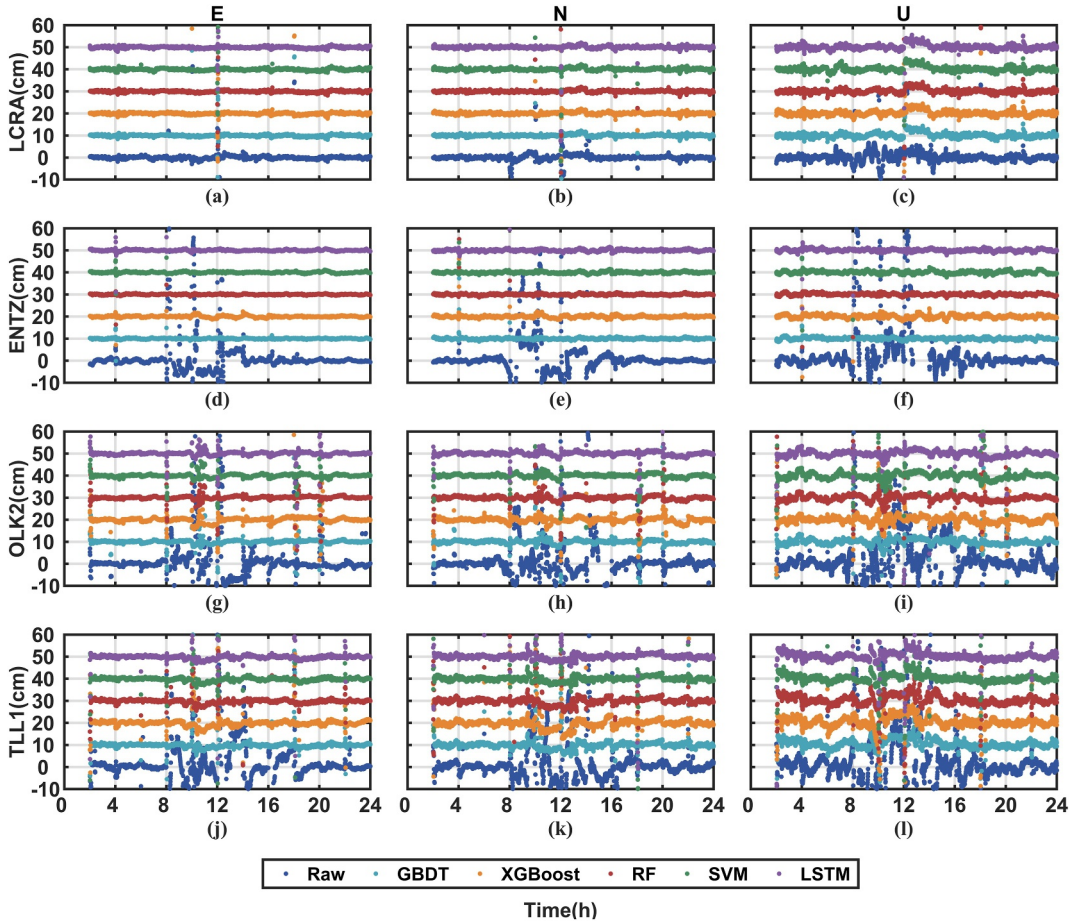


Figure 10. PPP-RTK positioning results with different ionospheric delay products.

derived from the traditional IDW method is insufficient to satisfy the requirements of high-precision PPP-RTK applications. In contrast, the machine learning-based ionospheric delay grid modeling method proposed in this study effectively improves the positioning accuracy at the user end. The most pronounced improvement is observed in the U direction, where the maximum RMSE reduction exceeds 60%. This clearly demonstrates the effectiveness of the proposed approach. Among the evaluated models, the RF method exhibits slightly better performance than the other methods in all three directions across multiple stations, indicating superior stability and robustness.

Convergence time is an important indicator for evaluating the performance of PPP-RTK. To comprehensively assess the performance of different ionospheric delay products, the first convergence time from multiple positioning sessions at the LCRA, ENTZ, OLK2, and TLL1 stations on DOY 001 in 2025 was statistically analyzed. The convergence time is defined as the first epoch at which the positioning errors in the E, N, and U components

Table 4
The Maximum Positioning Error of the PPP-RTK Solution

	Raw (cm)			GBDT (cm)			XGBoost (cm)			RF (cm)			SVM (cm)			LSTM (cm)		
Station	E	N	U	E	N	U	E	N	U	E	N	U	E	N	U	E	N	U
LCRA	44.5	23.6	79.0	38.5	28.2	29.7	38.4	28.5	27.7	38.7	28.1	29.2	38.9	28.2	31.9	38.9	28.0	29.5
ENTZ	99.5	49.9	126.6	4.9	27.3	5.9	2.5	26.2	4.5	4.4	25.1	1.5	6.7	28.5	6.6	5.9	22.4	3.7
OLK2	133.8	109.3	163.7	38.0	43.1	65.1	38.5	49.1	62.3	36.6	45.2	63.9	36.1	43.7	133.7	38.2	44.9	64.0
TLL1	90.3	137.0	83.2	75.3	67.8	54.7	75.5	65.1	54.9	75.3	66.7	72.0	75.4	56.9	62.7	75.5	56.5	64.1

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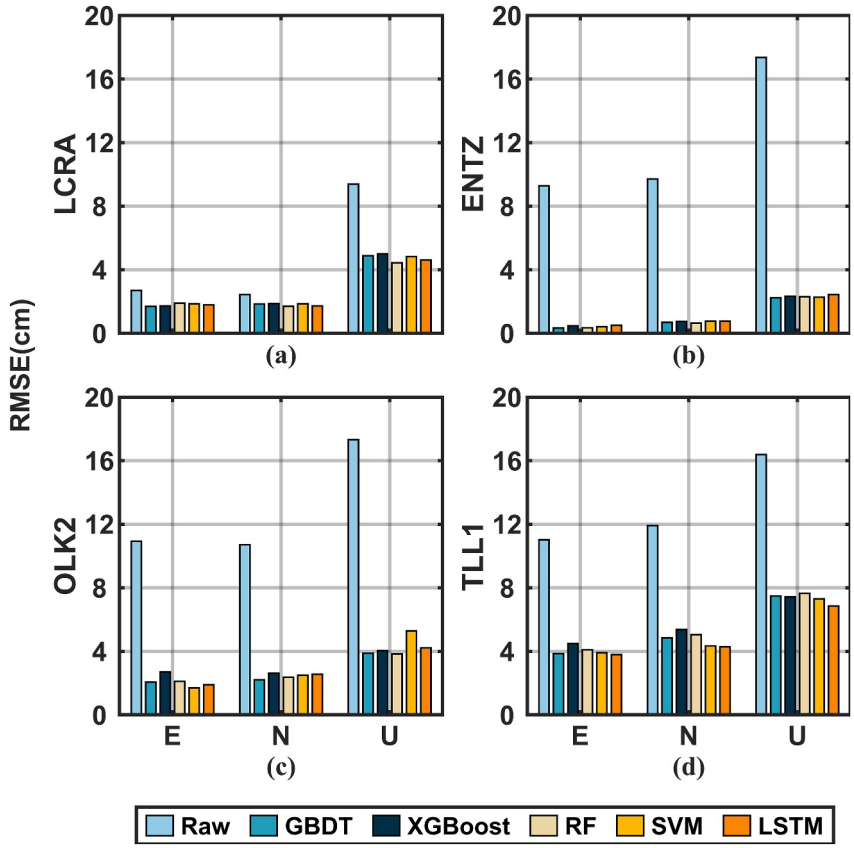


Figure 11. RMSE of PPP-RTK positioning for different stations.

remain within 10 cm for at least 10 consecutive epochs. The average convergence epochs for the experimental stations are listed in Table 5. As shown in the table, for the LCRA station, where the average inter-station distance is relatively small, the average convergence epoch obtained using the traditional IDW method is 3.55, fully demonstrating the rapid convergence capability of PPP-RTK. However, as the inter-station distance increases, the convergence time increases significantly. When the average inter-station distance reaches 260 km, the average convergence epoch increases to 27.27, which can no longer satisfy the requirements for real-time high-precision positioning. In contrast, the machine learning-based ionospheric delay grid modeling method effectively mitigates this problem. For the TLL1 station, the average convergence time is reduced by 74.7%, 72.0%, 73.3%, 75.0%, and 75.7%, respectively, compared with the IDW method. These results clearly demonstrate the effectiveness of the proposed approach in regions with sparse reference station networks.

To further validate the effectiveness of the proposed residual learning strategy, a direct machine learning-based approach, which directly establishes the mapping between the selected input features and ionospheric slant delays, is additionally employed for comparison. The two strategies are evaluated using identical data sets and experimental configurations to ensure a fair comparison. Figure 12 presents a comparison of the three-dimensional (3D) PPP-RTK positioning errors obtained using the direct machine learning and residual machine learning approaches for the five machine learning models at four stations. The comparison demonstrates the advantage of residual learning over direct learning for ionospheric delay modeling. For all five machine learning models, most of the samples are located above the reference line, indicating that the residual machine learning approach generally achieves lower 3D PPP-RTK positioning errors than the direct machine learning approach. The advantage is particularly evident at TLL1, where the direct machine learning approach exhibits substantially larger positioning errors, with numerous errors exceeding 10 cm. In contrast, the residual machine learning approach maintains relatively lower and more stable positioning errors. These results confirm that learning the ionospheric delay residuals, rather than directly learning the complete ionospheric delay, can effectively improve the accuracy and robustness of the ionospheric corrections used in PPP-RTK.

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Table 5
Average First Convergence Epoch of PPP-RTK for the Experimental Stations

Station	Raw	GBDT	XGBoost	RF	SVM	LSTM
LCRA	3.55	2.55	2.55	2.55	2.64	2.64
ENTZ	11.45	1.55	1.55	1.55	1.55	1.55
OLK2	22.91	5.91	7.18	6.36	5.36	5.27
TLL1	27.27	6.91	7.64	7.27	6.82	6.64

To demonstrate the general applicability of the machine learning-based PPP-RTK ionospheric delay grid modeling method proposed in this study, PPP-RTK positioning experiments were conducted using data from the LCRA, ENTZ, OLK2, and TLL1 stations from DOY 001 to 007 in 2025. The probabilities of positioning errors in the E, N, and U directions falling within different intervals (<2 cm, 2–5 cm, 5–10 cm, and 10–20 cm) were statistically analyzed. The results are presented in Figure 13.

For the LCRA station, the positioning errors are smaller than 2 cm in the horizontal direction and smaller than 5 cm in the vertical direction for most epochs, regardless of the method used. However, as the inter-station distance increases, the positioning accuracy obtained using the ionospheric delay products derived from the IDW method decreases significantly. For the TLL1 station, the proportion of epochs with horizontal positioning errors better than 2 cm is only about 60%, while the proportion with errors better than 5 cm is approximately 70%. In the vertical direction, the proportion of epochs with errors better than 5 cm is only 67.8%. These results indicate that the traditional interpolation-based approach is not suitable for ionospheric delay modeling in regions with

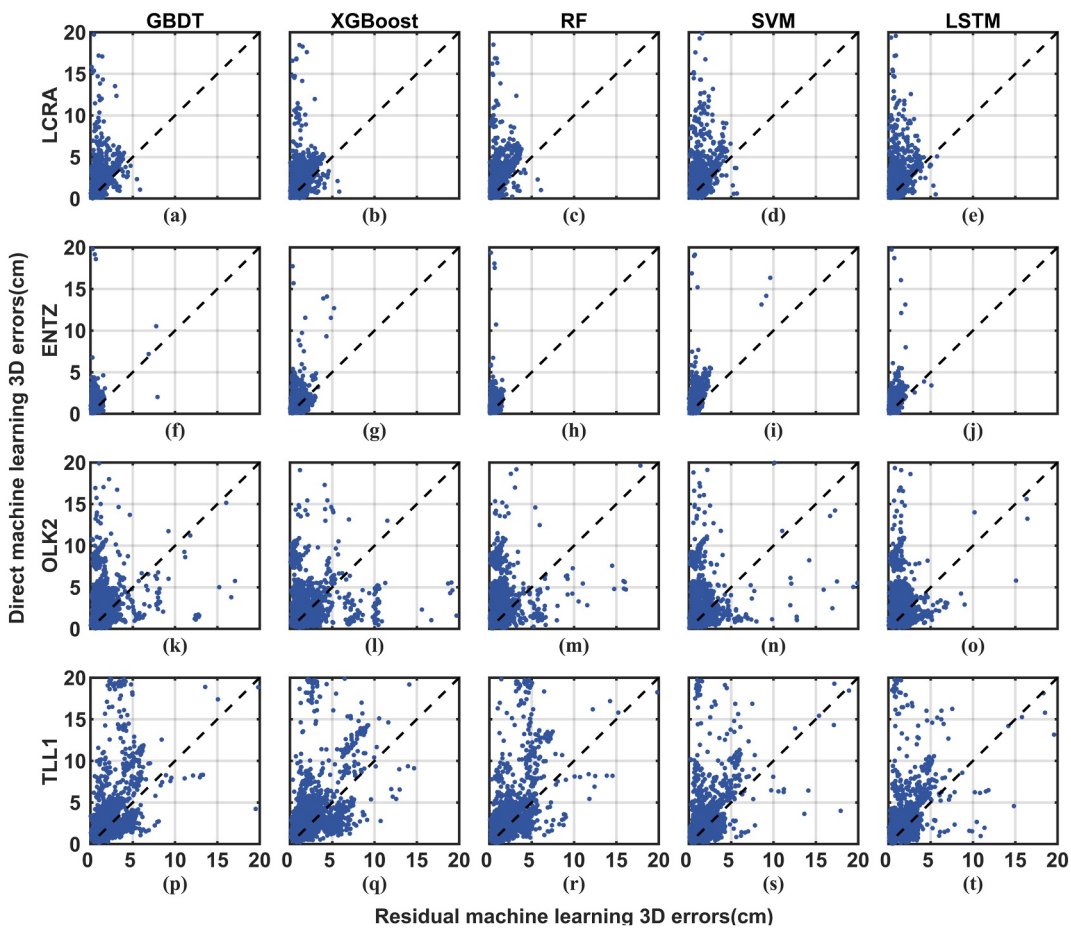


Figure 12. Comparison of 3D PPP-RTK positioning errors between direct and residual machine learning methods.

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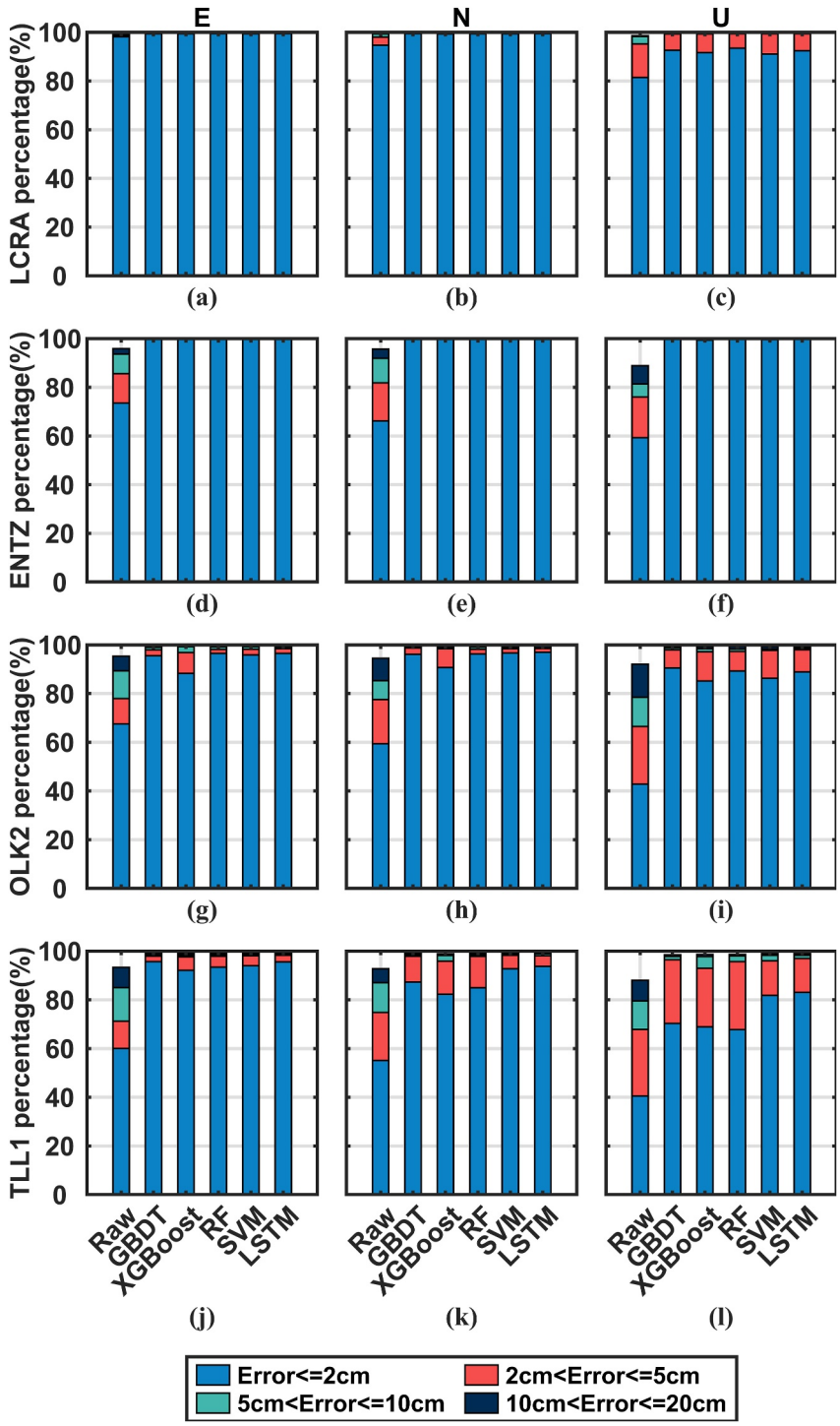


Figure 13. Probability distribution of positioning errors for experimental stations.

sparse reference station networks. In contrast, when the machine learning-based ionospheric delay grid modeling method is applied, more than 85% of the epochs at the TLL1 station still achieve positioning errors better than 2 cm in the horizontal direction and better than 5 cm in the vertical direction. These results further demonstrate the broad applicability of the proposed method. A comparison among different machine learning models shows that the RF model achieves slightly better positioning accuracy than the other methods in multiple experiments.

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Table 6
Convergence Time and 3D RMS of PPP-RTK Under Different Constraint Strategies

	Average first convergence epoch			3D RMS (cm)		
Station	Fix 5 cm	$k = 0.6$	$k = 1$	Fix 5 cm	$k = 0.6$	$k = 1$
LCRA	3.82	2.64	4.36	7.91	5.11	7.88
ENTZ	2.55	1.91	2.91	3.73	2.41	4.31
OLK2	4.18	3.36	3.55	6.51	4.98	5.87
TLL1	5.18	4.36	4.55	8.94	8.04	8.43

The constraint strategy adopted for the user-end ionospheric delay products has a significant impact on PPP-RTK positioning performance. To further evaluate the effectiveness of the uncertainty model proposed in this study, three ionospheric delay constraint strategies were investigated: a fixed 5 cm, an uncertainty-based constraint with a scaling factor of 0.6, and an uncertainty-based constraint with a scaling factor of 1.0. Since the RF model slightly outperformed the other machine learning models, only the RF model was considered in this experiment. 1-week data sets were used for the evaluation, and the average convergence time and three-dimensional RMS of the positioning results are summarized in Table 6. The experimental results indicate that the uncertainty-based constraint with a scaling factor of 0.6 slightly outperforms the other two strategies in terms of both convergence time and three-dimensional positioning accuracy, thereby demonstrating the effectiveness of the proposed uncertainty model. Nevertheless, it should be noted that the scaling factor should be determined according to the actual accuracy and statistical characteristics of the ionospheric delay products.

4.4. Performance Validation Under Severe Geomagnetic Disturbances

To investigate the robustness and applicability of the proposed machine learning-based PPP-RTK ionospheric delay grid modeling method under different geomagnetic storm conditions, data from the TLL1 station corresponding to various Kp index levels in 2025 were selected for experimental analysis. The Kp index data were obtained from the GFZ Helmholtz Centre for Geosciences. A total of 6 days of experimental data with Kp indices ranging from 2 to 8 were selected. The corresponding Kp indices and their associated geomagnetic storm levels are listed in Table 7. The trained machine learning models were then used to estimate the residuals of the ionospheric delay products, and the proportion of residuals smaller than 5 cm obtained by all methods was calculated. The corresponding statistical results are presented in Figure 14.

As shown in Figure 14, the proportion of residuals smaller than 5 cm for all methods decreases as geomagnetic disturbances intensify, indicating that geomagnetic activity significantly degrades the accuracy of ionospheric delay modeling. Even minor geomagnetic storms can noticeably affect PPP-RTK performance. Using the traditional IDW method based on spatial distance interpolation, the proportion of ionospheric delay errors smaller than 5 cm exceeds 90% during geomagnetically quiet periods but decreases to approximately 60% during geomagnetic storms, demonstrating the high sensitivity of conventional approaches to space weather disturbances. In contrast, the machine learning-based ionospheric delay grid modeling method proposed in this study maintains a proportion close to 80% for residuals smaller than 5 cm even under geomagnetic storm conditions, indicating that the proposed approach can effectively improve the robustness of PPP-RTK ionospheric delay

Table 7
The Daily Average of Kp Indices and Geomagnetic Storm Levels for Selected DOY in 2025

Doy	Kp index	Geomagnetic storm level
011	2.33	Quiet
020	3.67	Active
041	4.67	Active
123	5.33	Minor storm
258	6.67	Moderate storm
316	8.67	Major storm

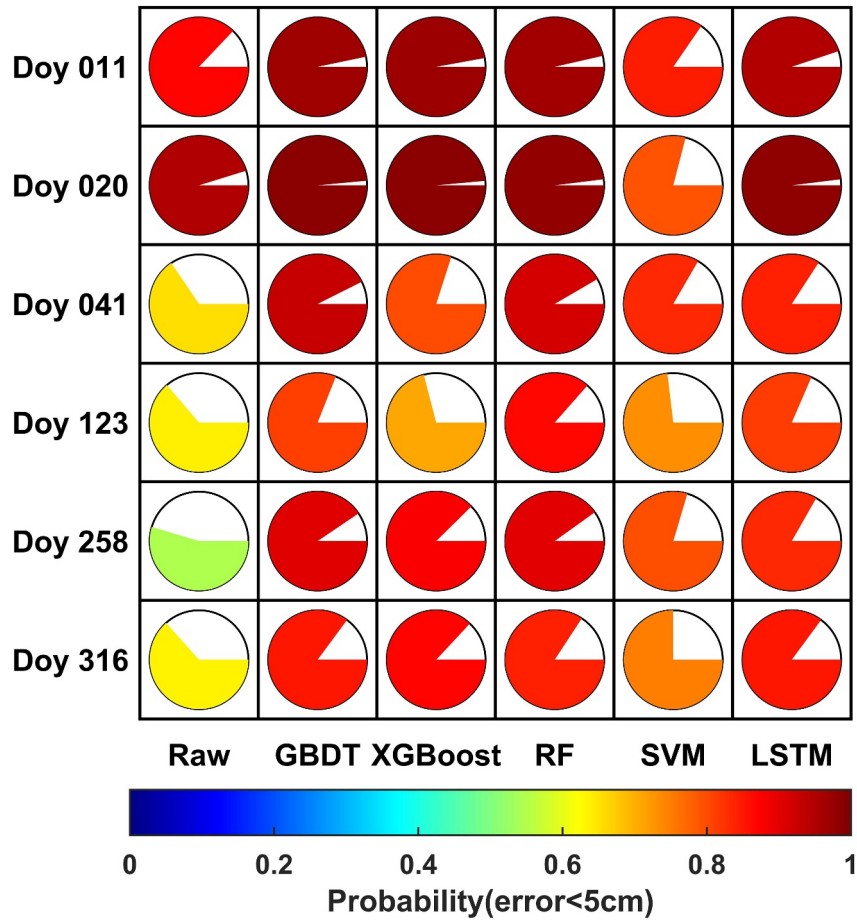


Figure 14. Probability of ionospheric delay residuals smaller than 5 cm.

modeling during disturbed geomagnetic conditions. Among the five machine learning models evaluated, the RF model shows the best overall performance. Even during major geomagnetic storms, the proportion of residuals smaller than 5 cm can still reach approximately 85%. These results demonstrate the clear advantage of machine learning-based ionospheric grid modeling under complex space weather conditions. Moreover, the RF model maintains relatively high accuracy during severe geomagnetic disturbances, making it a suitable choice for constructing robust ionospheric delay models.

Geomagnetic conditions can significantly affect the modeling accuracy of the ionosphere, thereby influencing the positioning performance of PPP-RTK at the user end. To further evaluate the reliability of the proposed method under different levels of geomagnetic disturbance, PPP-RTK positioning experiments were conducted using observation data from the TLL1 station on DOY 011, 020, 041, 123, 258, and 316 in 2025. As in the previous experiments, the observation data were artificially interrupted every 2 hrs to allow the positioning solutions to reconverge. Figure 15 presents the probability statistics of positioning errors smaller than 5 cm in the horizontal and vertical directions at the TLL1 station. As shown in the figure, when the traditional IDW method is used, the probabilities of horizontal and vertical positioning errors being smaller than 5 cm on DOY 316 are both below 60%. In contrast, the machine learning-based ionospheric delay grid modeling method significantly improves the positioning performance at the user end during geomagnetic storm conditions. Even under major geomagnetic storms, the probabilities of horizontal and vertical positioning errors being smaller than 5 cm still exceed 90%, clearly demonstrating the robustness of the proposed method under extreme space weather conditions. Furthermore, comparisons across multiple experiments indicate that the RF model performs slightly better than the other machine learning models.

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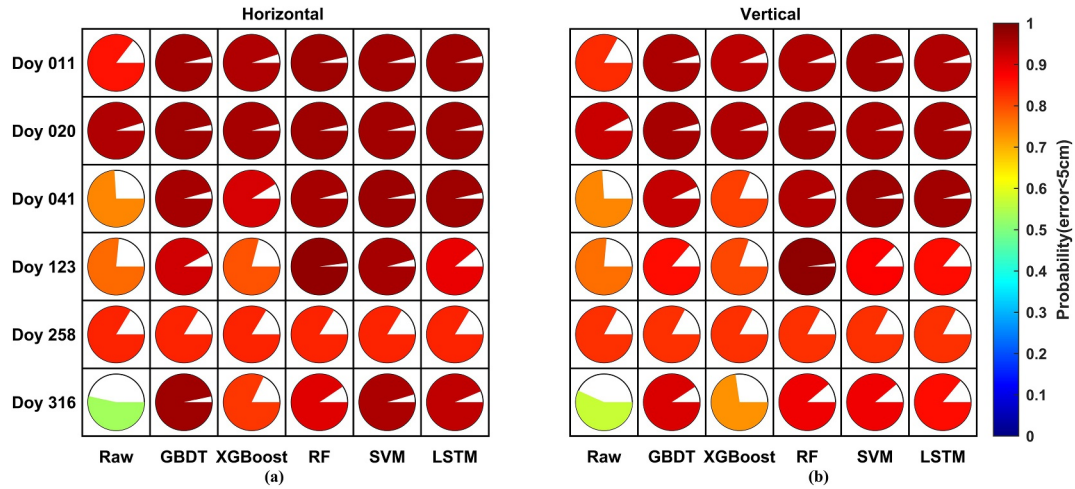


Figure 15. Probability of horizontal and vertical errors less than 5 cm at TLL1 under different geomagnetic disturbance levels.

5. Conclusions

Sparse reference station networks and severe geomagnetic disturbances can degrade the accuracy of ionospheric delay modeling at the network side, thereby reducing the positioning performance of PPP-RTK at the user end. To address this issue, this study proposes a machine learning-based ionospheric delay grid modeling method. By finely modeling the ionospheric delay computation errors and interpolation model errors embedded in the ionospheric delay residuals, the robustness of PPP-RTK in practical applications is significantly improved.

To evaluate the performance of the proposed method, 274 stations uniformly distributed across Europe were used for experiments, among which 70 stations were selected as rover stations. Observation data from the entire year of 2024 were used for machine learning model training, while partial observations from 2025 were used for validation. Five machine learning models, including GBDT, XGBoost, RF, SVM, and LSTM, were comparatively analyzed. The experimental results show that for the TLL1 station, with an average reference station spacing of 260 km, the ionospheric delay products generated by the proposed method exhibit errors smaller than 10 cm for most epochs. Compared with the traditional IDW method, the five machine learning models improve the average RMSE by 60.2%, 57.1%, 61.6%, 49.3%, and 50.7%, respectively, while maintaining mean values close to zero, indicating unbiased estimation. Statistical results from all 70 stations show that the proportion of regional ionospheric delay modeling errors exceeding 6 cm is 22% when using the IDW method, whereas it decreases to only 5% when using the RF-based model. Furthermore, during severe geomagnetic storms, the probability that the ionospheric delay error remains smaller than 5 cm exceeds 85% when using the RF-based modeling method, whereas it decreases to about 60% when using the IDW approach. These results indicate that the proposed method significantly improves the robustness of ionospheric delay modeling under disturbed space weather conditions.

PPP-RTK positioning experiments further demonstrate that the proposed method enables rapid or even instantaneous convergence in regions with sparse reference station networks, while maintaining continuous and stable positioning results. For the TLL1 station, the maximum RMSE in the U direction is reduced by more than 60%. The average convergence time is reduced by 74.7%, 72.0%, 73.3%, 75.0%, and 75.7% for the five machine learning models compared with the IDW method. 1-week experimental results show that the probability of achieving positioning errors better than 2 cm in the horizontal direction and 5 cm in the vertical direction exceeds 85% when using the proposed method, whereas it is only 67.8% when using the IDW method. Moreover, under severe geomagnetic storm conditions, the probability that both horizontal and vertical positioning errors are smaller than 5 cm remains above 90%, while it is below 60% when using the IDW method.

Overall, the experimental results demonstrate the effectiveness and general applicability of the proposed machine learning-based ionospheric delay grid modeling method in regions with sparse reference station networks and under severe geomagnetic disturbances, providing a practical solution for improving the robustness of PPP-RTK positioning.

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Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

The GNSS observation data used in this study are publicly available from the EUREF Permanent GNSS Network (EPN) data centres at https://epncb.oma.be/ (Bruyninx et al., 2019). The precise orbit, clock, and OSB products are publicly available from the Wuhan University IGS Analysis Center at http://www.igs.gnsswhu.cn (Jiang et al., 2023).

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